TED (21) – 2041 REVISION 2021

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY
(Common to BM / EC / EL)

BASIC ELECTRONICS CIRCUITS

MODEL QUESTION PAPER – SET-1

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	An intrinsic semiconductor at absolute zero temperature behaves like	M 1.01	U
2	Name a trivalent impurity used in extrinsic semiconductor.	M 1.01	R
3	Write the equation to calculate the dynamic resistance of a forward biased diode.	M 1.03	R
4	In a transistor .... region is larger in size and medium in doping.	M 2.01	R
5	Draw symbol of PNP transistor and mark its terminals.	M 2.01	R
6	FET is a .... controlled device.	M3.01	R
7	Draw the equivalent circuit of a UJT.	M3.01	R
8	State the relationship between Xc and R in a good RC differentiator circuit.	M 4.03	R
9	State the need of filter in rectifier circuit.	M 4.02	U

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Differentiate drift current and diffusion current.	M 1.02	U
2	Define the terms forward voltage drop, maximum forward current, and peak inverse voltage of PN diode.	M 1.04	R
3	Define reverse saturation current in PN diode.	M 1.03	R
4	Define barrier potential in PN junction.	M 1.02	R

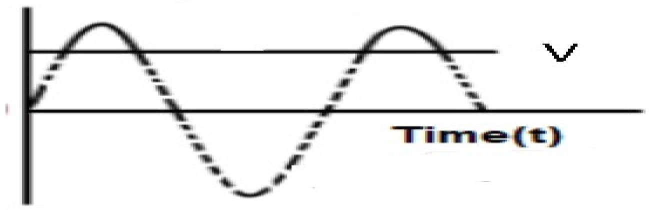
5	Compare emitter, base and collector regions in a BJT.	M 2.01	R
6	Derive the expression for □ in terms of □.	M 2.03	U
7	Explain the transistor action.	M 2.05	U
8	State why CE configuration is widely used in amplifiers.	M 2.04	R
9	Draw VI characteristics of UJT and mark operating regions.	M 3.03	R
10	Differentiate linear and non linear wave shaping circuits.	M 3.03	U

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Compare intrinsic and extrinsic semiconductor.	M 1.01	R
	OR		
IV	Draw the atomic structure and explain the formation of N type semiconductor.	M 1.01	U
V	With the help of a sketch explain the operation of NPN transistor.	M 2.02	U
	OR		
VI	Draw output characteristics of NPN transistor in CE configuration, and explain various operating regions.	M 2.04	U
VII	Draw structure of N channel JFET and explain its principle of operation.	M 3.01	U
	OR		
VIII	Draw the drain characteristics of N channel depletion type MOSFET. Compare MOSFET and JFET.	M 3.03	R
IX	Draw drain characteristics of N channel JFET, and discuss various operating regions.	M 3.03	U
	OR		
X	Draw the structure and explain the working of N channel depletion type MOSFET	M 3.02	U
XI	Design a diode circuit to develop an output as shown, and explain operation. Assume the diode is ideal, and suitable sine wave as	M 4.04	A

	input.  		
	OR		
XII	Design a diode circuit to shift the base of the sine wave input VmSineωt to –Vm, and explain operation	M 4.04	A
XIII	Draw circuit diagram and explain operation of RC differentiator.	M 4.03	U
	OR		
XIV	Draw a voltage quadrupler circuit, and explain its operation.	M 4.05	U

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SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY
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BASIC ELECTRONICS CIRCUITS

MODEL QUESTION PAPER – SET-2

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	Name any donor type impurity used in extrinsic semiconductor.	M 1.01	U
2	The width of depletion layer within a PN junction increase with .... bias.	M 1.02	R
3	.... configuration offers highest voltage gain.	M 2.04	R
4	In saturation mode of operation, collector base junction has .... bias.	M 2.02	R
5	A field effect transistor operates using .... carriers only.	M 3.01	R
6	Operation of a N channel depletion type MOSFET needs a .... gate voltage.	M 3.02	R
7	Draw the symbol of N channel MOSFET	M 3.01	R
8	State equation for calculating DC output voltage in a centre tap full wave rectifier.	M 4.01	R
9	.... is the output of an integrator when the input is rectangular wave.	M 4.03	U

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	List the features of intrinsic semiconductor.	M 1.01	R
2	Draw the energy band diagram of conductor, semiconductor and insulator.	M 1.03	R

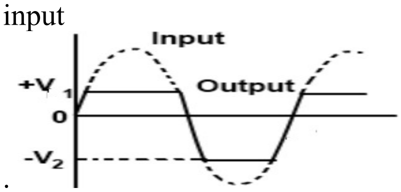
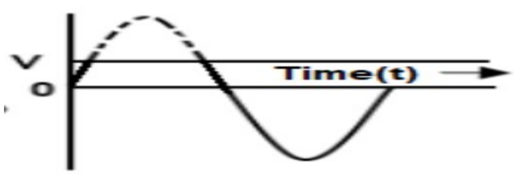
3	Draw forward VI curve of a diode and show how to calculate static and dynamic resistance.	M 1.03	R
4	Explain zener breakdown.	M 1.03	U
5	State biasing conditions for active, saturation and cut-off modes of operation in a BJT.	M 2.02	R
6	Draw transfer characteristics of N channel depletion type MOSFET, and mark various regions.	M 3.03	R
7	Write note on pinch off voltage of JFET.	M 3.02	U
8	Define dynamic drain resistance, trans-conductance, and amplification factor for JFET.	M 3.03	R
9	Draw transfer curve of N channel JFET. Write expression for drain current in terms of maximum drain current and pinch off voltage.	M 3.03	R
10	Compare capacitor and inductor type filters used in power supplies.	M4.02	R

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

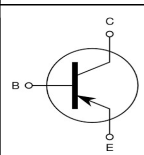
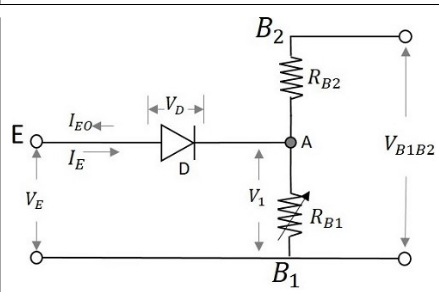
III	Draw the atomic structure and explain the formation of P type semiconductor.	M 1.01	U
	OR		
IV	Draw a sketch and explain the formation of PN junction and depletion region.	M 1.02	U
V	Draw the symbol of NPN and PNP transistor. Draw structure of NPN transistor and explain features of different sections.	M 2.01	U
	OR		
VI	Draw output characteristics of NPN transistor in CB configuration, and explain various operating regions.	M 2.04	U
VII	Draw input characteristics of NPN transistor in CE configuration, and explain the behaviour. Show how input resistance is calculated.	M 2.04	U
	OR		
VIII	Draw sketch and explain mechanism of current flow and current relations in NPN transistor.	M 2.02	U

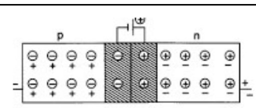
<table>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>IX</td><td>Compare JFET and BJT.</td><td>M 3.04</td><td>R</td></tr>
<tr><td></td><td>OR</td><td></td><td></td></tr>
<tr><td>X</td><td>Draw structure and explain working of UJT.</td><td>M 3.02</td><td>U</td></tr>
<tr><td>XI</td><td>Draw circuit diagram and explain operation of RC integrator.</td><td>M 4.03</td><td>U</td></tr>
<tr><td></td><td>OR</td><td></td><td></td></tr>
<tr><td>XII</td><td>Draw circuits and explain operation of half wave and full wave voltage doubler.</td><td>M 4.04</td><td>U</td></tr>
<tr><td>XIII</td><td>Design a diode circuit to develop an output as shown, and explain operation. Assume the diode is ideal, and suitable sine wave as input

</td><td>M 4.04</td><td>A</td></tr>
<tr><td></td><td>OR</td><td></td><td></td></tr>
<tr><td>XIV</td><td>Design a diode circuit to develop an output as shown, and explain operation. Assume the diode is ideal, and suitable sine wave as input.

</td><td>M 4.04</td><td>A</td></tr>
</table>

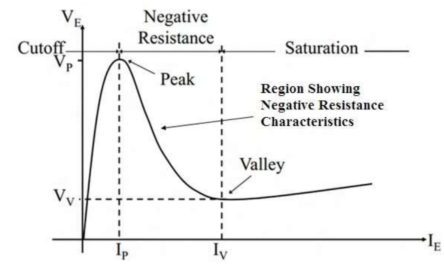
Scoring Indicators

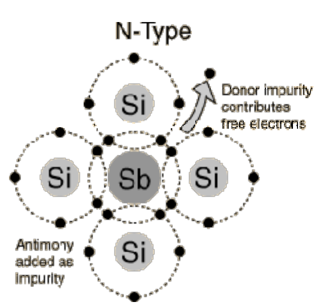
Model Question Paper Set 1

BASIC ELECTRONIC CIRCUITS

Q No	Scoring Indicators	Split score	Sub Total	Total score
	PART A			**9**
I. 1	Insulator		1	
I. 2	Gallium, indium, aluminium, boron(any one)		1	
I. 3	Rd=□Vf/□□f or change in forward voltage/change in current		1	
I. 4	Collector		1	
I. 5			1	
I. 6	Voltage		1	
I. 7			1	
I. 8	Xc>>R		1	
I. 9	To remove ripple from rectifier output.		1	
	PART B			24
II. 1	Diffusion current- due to movement of carriers from a high density area to low density area in a semiconductor without	1.5	3	

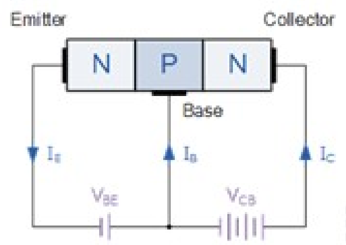
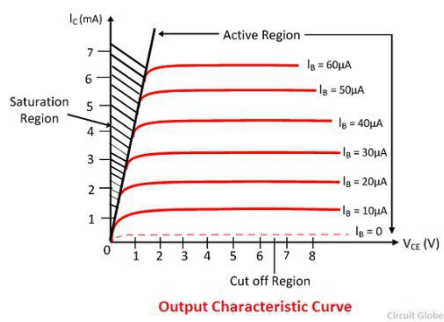
	applying external potential. Drift current – due to movement of carriers inside semiconductor under influence of external field.	1.5		
II. 2	Forward voltage drop, V_F – The maximum voltage drop across diode, for a given forward current and device temperature. Maximum Forward Current $I_{F(max)}$ - is the maximum forward current allowed to flow through the device, without damaging. Peak Inverse Voltage (PIV) - is the maximum allowable Reverse operating voltage that can be applied across the diode without damage occurring to the device.	1*3	3	
II. 3	When the diode is reverse biased then the depletion region width increases, majority carriers move away from the junction and there is no flow of current due to majority carriers. But thermally produced electron hole pairs will cause current to flow in the circuit. This current is usually very small (in terms of micro amp to nano amp). Since the current is almost constant known as reverse saturation current I_{CO}.	3	3	
II. 4	 When PN junction is formed, immobile +ve ions appears near junction in N side and _ve ions on P side. This layers of charges creates field which prevents further movement of charges across junction. This is barrier potential ; about .6v for Si.	3	3	
II. 5	Emitter – supplies carriers, heavily doped , medium size. Base – passage between emitter and collector, small, light doping. Collector – collects the carriers, large, medium doping.	3	3	
II. 6	Ie=Ib+Ic; Ie, Ie/Ic=Ib/Ic+Ic/Ic=➔1/□=1/□+1 1/□=(1+□)/□ Ie, □=□□ /(□□ +1)	3	3	
II. 7	A transistor transfers the input signal current from a low-resistance circuit (forward biased EB junction)to a high-resistance circuit(reverse biased CB junction).- Ie=Ic. This is the key factor responsible for the amplifying capability of the transistor.	3	3	
II. 8	The CE configuration provides both High Current and Voltage gain unlike other configurations like CC (High current gain but	3	3	

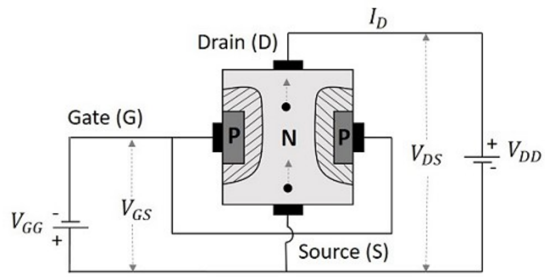
<table>
<tr><td></td><td>voltage gain less than unity i.e 1) and CB (High voltage gain but current gain less than unity). i/p and o/p resistance are also favorable.</td><td></td><td></td><td></td></tr>
<tr><td>II.9</td><td>
</td><td>3</td><td>3</td><td></td></tr>
<tr><td>II.10</td><td>In the non-linear circuit, the non-linear devices like diodes are used, not have any linear relationship between the current & voltage

In the linear circuits, the linear element like resistor, capacitor and inductance are used and there will be a linear relationship between the voltage and current.</td><td></td><td>3</td><td></td></tr>
<tr><td></td><td>PART C</td><td></td><td></td><td>42</td></tr>
<tr><td>III</td><td><table>
<tr><th>Intrinsic</th><th>Extrinsic</th></tr>
<tr><td>Pure semiconductor</td><td>Added with impurity</td></tr>
<tr><td>Low conductivity</td><td>High conductivity</td></tr>
<tr><td>Temperature dependent conductivity</td><td>Almost temp independent operation</td></tr>
<tr><td>Thermally generated carriers only</td><td>Added carriers</td></tr>
<tr><td>Minority carrier only</td><td>Minority and majority carriers</td></tr>
<tr><td>Not much used in devices</td><td>Used in semiconductor devices</td></tr>
<tr><td>Equal no of electrons and holes</td><td>Electron hole nos are different</td></tr>
<tr><td>Fermi level is present in the middle of forbidden energy gap.</td><td>The presence of Fermi level varies according to the type of extrinsic semiconductor.</td></tr>
<tr><td>Energy gap is small.</td><td>The energy gap is more than that in an intrinsic semiconductor.</td></tr>
</table>
(any 7 points)</td><td>7*1</td><td>7</td><td>7</td></tr>
<tr><td>IV</td><td>

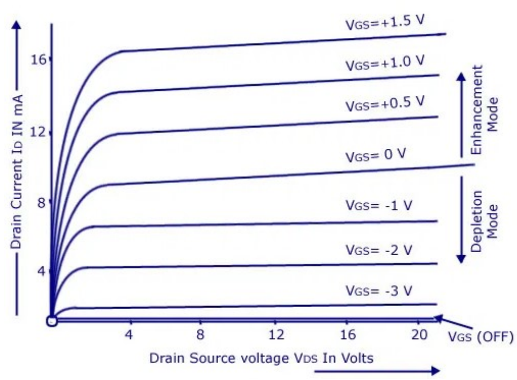
In intrinsic semiconductor crystal, all atom forms covalent bonds with four adjacent atoms, no free electrons, acting as</td><td>3

4</td><td>7</td><td>7</td></tr>
</table>

	insulator at absolute zero. Impurity atoms with 5 valence electrons when added, they form bonds with four atoms by sharing four outer band electrons, but fifth one pushed to next free level, where it is free to move. Every impurity atom donates one conducting electron, produce n-type semiconductors.			
V	 E-B junction forward and C-B reverse. Electrons entering emitter causes Ie flowing out. These electrons crossing junction, and few recombines with +ve carriers in base causing base current Ib. Only few electrons recombines as base width is less and doping is low, so Ib is very low. Majority of electrons from emitter crossing B-C junction and reaching collector, going to +ve terminal of Vcb causing collector current Ic. Collector current almost same as emitter current.	3 4	7	7
VI	 plot of the collector current, I_C, versus the collector-emitter voltage, V_{CE}, for various values of the base current, I_B. When Ib=0, cut off region, Ic is vary small (leakage current) When Vce increases from zero collector current increases rapidly, Ic=□□b.-saturation region. Ic saturates around Vce=1V, and no much increase in Ic with Vce. Active region.	3 4	7	7

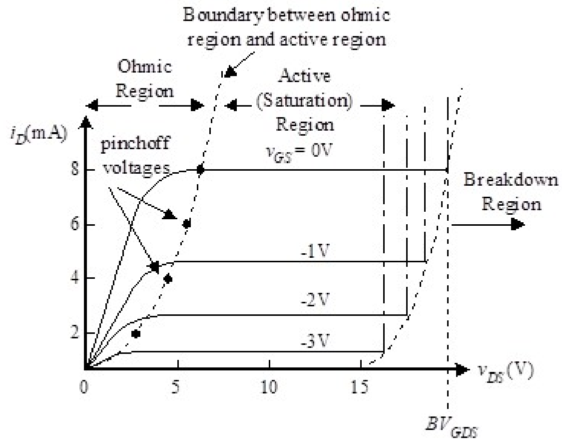
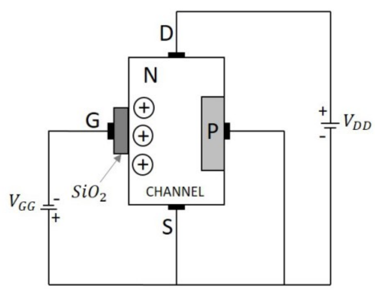
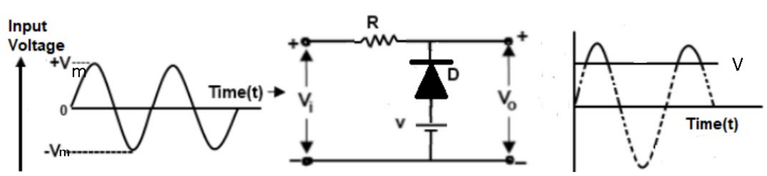
<table>
<tr><td>VII</td><td>

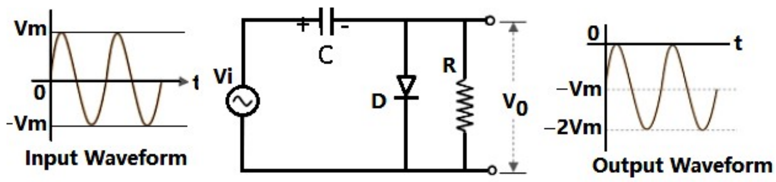
PN junction between gate and channel; depletion region more in channel due to less doping. Drain +ve with source, andno gate voltage, drain current flows through channel. –ve gate voltage widens depletion region, drain current reduces due to narrow channel.for a set drain voltage, drain durrent is function of gate –ve voltage. JFET- voltage controlled device. The gate voltage at which drain current becomes zero is pinch off voltage.</td><td>3

4</td><td>7</td><td>7</td></tr>
<tr><td>VIII</td><td>

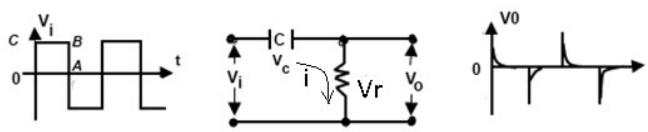
<table>
<tr><th></th><th>FET</th><th>MOSFET</th></tr>
<tr><td>Operational modes</td><td>Depletion mode only</td><td>Both Enhancement and Depletion modes</td></tr>
<tr><td>Input impedance</td><td>High</td><td>Very high</td></tr>
<tr><td>Output resistance</td><td>Moderate</td><td>Low</td></tr>
<tr><td>Operational speed</td><td>Moderate</td><td>High</td></tr>
<tr><td>Thermal stability</td><td>Better</td><td>High</td></tr>
</table>
(any three points)</td><td>4

3</td><td>7</td><td>7</td></tr>
</table>

IX	 Plot between Id and Vds for different Vgs. Vg & Vds=0 no Id. As Vds increases, Id increase initially- (ohmic region). Increase in Vds increases reverse bias between the gate drain PN junction, increases width of depletion region in channel, restricts channel width, Id restricted even with rise in Vds.(pinch off/saturation region). At very high Vds, breakdown occurs in channel, damaging channel(break down region). Increase in –ve Vgs reduces channel width, and Id falls for same Vds	4 3	7	7
X	 Working of MOSFET in depletion mode No gate voltage, drain current flows due to Vds. In depletion mode, negative gate potential repels electrons in channel, increases its resistance, reducing drain current. At Vgg(off), channel shuts off and drain current becomes zero. Enhancement mode - With +ve gate voltage, more electrons induced in to the channel, and drain current increases with increase in Vgg.	3 4	7	7
XI	 Biased –ve clipper. During –ve half cycle, and in +ve half cycle for Vi<V, diode conducts, and o/p same as V. for Vi>V, diode reverse biased, o/p same as Vi.	3 4	7	7

<table>
<tr><td>XII</td><td>


-ve clamper circuit. During C charges to Vm through D. capacitor voltage Vm added with i/p voltage, like a battery, to shift o/p reference to –Vm.
</td><td>4

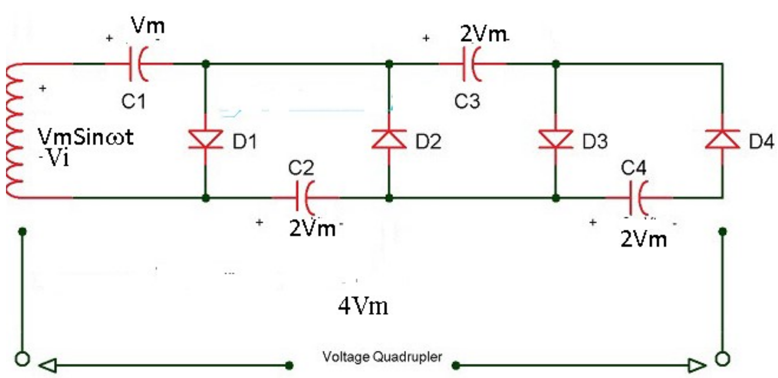
3</td><td>7</td><td>7</td></tr>
<tr><td>XIII</td><td>


RC differentiator. Vi=Vc+Vr. o/p Vo is voltage across R, ie Vr.
Circuit current i=Vc/Xc=Vr/R.
If Xc>>R, Vc=Vi.
i=q/t; dq/dt. Q=Vc*C. ie

$i = C\frac{d}{dt}(V_c)$ or $i = C\frac{d}{dt}(V_i)$

Out put Vo= i*R. ie o/p voltage proportional to the integral of Vi

$= RC\frac{d}{dt}(V_i)$; provided Xc>>R, and RC<<period of input.
</td><td>2

5</td><td>7</td><td>7</td></tr>
<tr><td>XIV</td><td>


During the positive cycle, diode D1and D3 forward biased and , C1 charges to Vm. diodes D2 and D4 are reverse biased.

During the negative cycle, through D2, C2 charges to 2Vm, as C1 voltage added with input voltage.

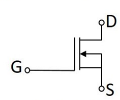
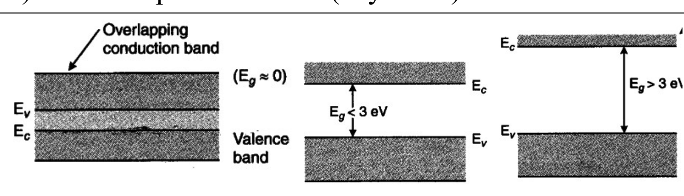
Similarly, C3 charges to 2Vm through D3 (Vi-Vm+2Vm) during +ve half, and C4 to 2Vm during –ve half cycle. Voltage across C2and C4 is 2Vm+2Vm=4Vm
</td><td>3

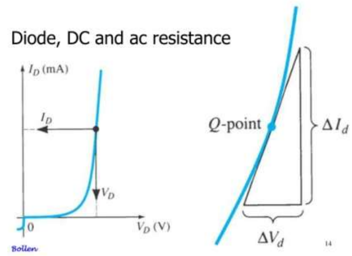
4</td><td>7</td><td>7</td></tr>
</table>

Scoring Indicators

Model Question Paper II

BASIC ELECTRONIC CIRCUITS

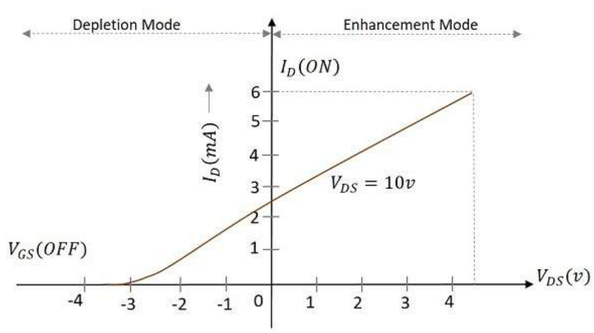
Q No	Scoring Indicators	Split score	Sub Total	Total Score
	PART A			**9**
I. 1	Phosphorus, antimony, arsenic, bismuth *(Write any one)*		1	1
I. 2	Reverse bias		1	1
I. 3	Common emitter		1	1
I. 4	Forward bias		1	1
I. 5	Majority carriers.		1	1
I. 6	Negative.		1	1
I. 7	N-Channel MOSFET  Depletion Mode		1	1
I. 8	2Vm/π		1	1
I. 9	Saw tooth		1	1
	PART-B			24
II. 1	1)materials in pure form show the property of semiconductor are called intrinsic semiconductor. 2)the number of free electrons in the conduction band is equal to the number of holes in the valence bond. 3)Its electrical conductivity is low. 4)Its electrical conductivity depends on temperature only. 5)It is of no practical use. (any three)	1*3	3	3
II. 2	 Conductor semiconductor insulator	1*3	3	3

<table>
<tr><td>II. 3</td><td>

Dyanamic resistance:

rd= ΔVd/ΔId

Static resistance:

Rd= Vd/Id</td><td>3 *1</td><td>3</td><td>3</td></tr>
<tr><td>II. 4</td><td>When the high electric-field is applied across the PN junction diode, under reverse bias, then the electrons start flowing across the PN-junction, and current flows in reverse bias. Happens in diodes with thin depletion region. Generally not damaging the junction.</td><td>3</td><td>3</td><td>3</td></tr>
<tr><td>II. 5</td><td><table>
<tr><th>Modes</th><th>Emitter-Base junction</th><th>Collector- Base junction</th></tr>
<tr><td>Cutoff</td><td>Reverse</td><td>Reverse</td></tr>
<tr><td>Active</td><td>Forward</td><td>Reverse</td></tr>
<tr><td>Saturation</td><td>Forward</td><td>Forward</td></tr>
</table></td><td>1*3</td><td>3</td><td>3</td></tr>
<tr><td>II. 6</td><td>

Transfer Characteristics of a MOSFET</td><td>3</td><td>3</td><td>3</td></tr>
<tr><td>II.7</td><td>For VGS=0 V, the value of VDS at which ID becomes essentially constant is the pinch-off voltage, VP. For a given JFET, VP has a fixed value. A continued increase in VDS above the pinch-off voltage produces an almost constant drain current. the drain current is a function of reverse bias voltage at gate.</td><td>3</td><td>3</td><td>3</td></tr>
</table>

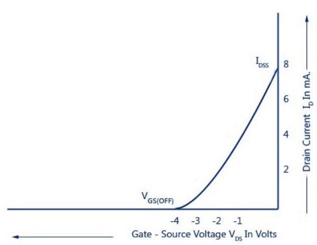
<table>
<tr><td>II. 8</td><td>This is the ratio of change of drain to source voltage (δV_{DS}) to the change of drain current (δI_D) at a constant gate to source voltage (V_{GS} = Constant). The ratio is denoted as r_d.

$$r_d = \frac{\delta V_{DS}}{\delta I_D} \quad at\ constant\ \ V_{GS}$$

Transconductance is the ratio of change in drain current (δI_D) to change in the gate to source voltage (δV_{GS}) at a constant drain to source voltage (V_{DS} = Constant).

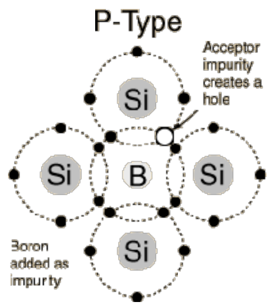
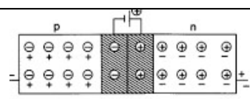
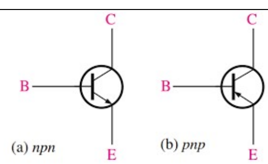
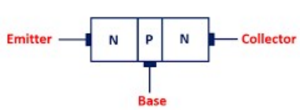
$$g_m = \frac{\delta I_D}{\delta V_{GS}} \quad at\ constant\ \ V_{DS}$$

The amplification factor is defined as the ratio of change of drain voltage (δV_{DS}) to change of gate voltage (δV_{GS}) at a constant drain current (I_D = Constant).

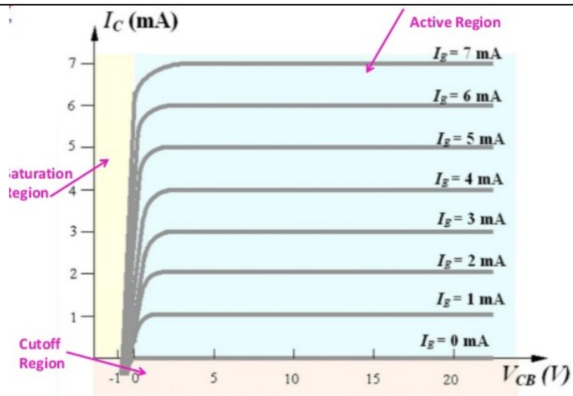
$$\mu = \frac{\delta V_{DS}}{\delta V_{GS}} \quad at\ constant\ \ I_D$$</td><td>1+1+1</td><td>3</td><td>3</td></tr>
<tr><td>II. 9</td><td>

$$I_D = I_{DSS}\left[1-\frac{V_{GS}}{V_P}\right]^2$$</td><td>2

1</td><td>3</td><td>3</td></tr>
<tr><td>II.10</td><td><table>
<tr><th>Parameter</th><th>Capacitor / C Filter</th><th>Choke input /L filter</th></tr>
<tr><td>Place of filter</td><td>Across the load</td><td>In series with the load</td></tr>
<tr><td>Useful in</td><td>Reducing ripple in load voltage</td><td>Reducing ripple in load current</td></tr>
<tr><td>Lowest ripple in the load voltage</td><td>No load or light loads</td><td>Heavy loads</td></tr>
<tr><td>Suitable for</td><td>Light load applications</td><td>Heavy load applications</td></tr>
<tr><td>Surge current through diodes</td><td>Very high and must be controlled</td><td>Low and need not to be controlled</td></tr>
<tr><td>Expression for ripple factor</td><td>RF= 1/4√3fCR</td><td>RF=R/3√2∞L</td></tr>
<tr><td>Size of filter</td><td>Small and compact</td><td>Bulky</td></tr>
<tr><td>Cost</td><td>Lowest</td><td>Moderate</td></tr>
<tr><td>Voltage regulation</td><td>Poor</td><td>Moderate</td></tr>
<tr><td>Applications</td><td>Cell charger, small eliminators</td><td>High current dc power supplies</td></tr>
</table>
(three points)</td><td>3</td><td>3</td><td>3</td></tr>
<tr><td></td><td>PART-C</td><td></td><td></td><td>42</td></tr>
</table>

III	 In intrinsic semiconductor crystal, all atom forms covalent bonds with four adjacent atoms, no free electrons, acting as insulator at absolute zero. Impurity atoms with 3 valence electrons when added, they form bonds with four atoms by sharing three outer band electrons, and one bone has absence of electron. It has ability to accept one electron, like a mobile +ve charge, or known as hole. Every impurity creates one hole, produce P-type semiconductors.	3 4	7	7
IV	 When the N-type semiconductor and P-type semiconductor joined together free electrons from N side crosses to P side and recombines with holes. Exposing immobile +ve ions in N side, and _ve ions in P side near junction. The field thus created prevents further flow of carriers- potential barrier. This region called depletion region- with no carriers. Forward bias reduces, reverse bias increases width.	3 4	7	7
V	  Emitter – supplies carriers, heavily doped , medium size. Base – passage between emitter and collector, small, light doping.	2	7	7

<table>
<tr><td></td><td>Collector – collects the carriers, large, medium doping.</td><td>2

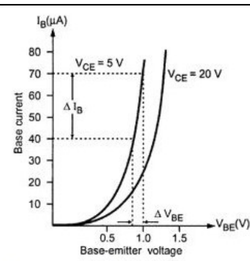
3</td><td></td><td></td></tr>
<tr><td>VI</td><td>

plot of the collector current, I_C, versus the collector-base voltage, V_{Cb}, for various values of the base current, $I_{e.}$

in saturation region V_{cb} is –ve, both junction forward biased, collector current rises with Vcb.

Vcb more than zero, collector junction reverse biased, Ic=Ie – i/p current (active region).

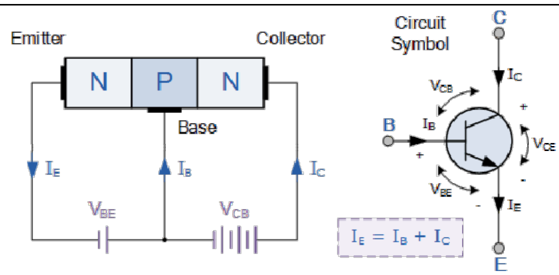
For Ie=0, cut off region, Ic = leakage current.</td><td>3

4</td><td>7</td><td>7</td></tr>
<tr><td>VII</td><td>

Input characteristics of the transistor in CE configuration

Plot between i/p current Ib against i/p voltage Vbe, for constant value of o/p voltage Vce. Like diode, no current up to knee voltage ,about .6 for silicon, there after increases rapidly. Ib very low as i/p impedance very high. With high Vce, Ib low as width of depletion region increases in Base region.</td><td>3

3</td><td>7</td><td>7</td></tr>
</table>

<table>
<tr><td></td><td>i/p resistance ri=□Vbe/□Ib</td><td>1</td><td></td><td></td></tr>
<tr><td>VIII</td><td>

E-B junction forward and C-B reverse. Electrons entering emitter causes Ie flowing out. These electrons crossing junction, and few recombines with +ve carriers in base causing base current Ib. Majority of electrons from emitter crossing base and reaching collector, going to +ve terminal of Vcb causing collector current Ic.

So Ie=Ib+Ic. Ib is <<Ic. so Ie=Ic.</td><td>3

4</td><td>7</td><td>7</td></tr>
</table>

<table>
<tr>
<td>IX</td>
<td>
<table>
<tr><th>BJT</th><th>FET</th></tr>
<tr><td>BJT stands for bipolar junction transistor, so it is a bipolar component</td><td>FET stands for the field-effect transistor, so it is a uni-junction transistor</td></tr>
<tr><td>BJT has three terminals like base, emitter, and collector</td><td>FET has three terminals like Drain, Source, and Gate</td></tr>
<tr><td>The operation of BJT mainly depends on both the charge carriers like majority as well as minority</td><td>The operation of FET mainly depends on the majority charge carriers either holes or electrons</td></tr>
<tr><td>The input impedance of this BJT ranges from 1K to 3K, so it is very less</td><td>The input impedance of FET is very large</td></tr>
<tr><td>BJT is the current controlled device</td><td>FET is the voltage-controlled device</td></tr>
<tr><td>BJT has noise</td><td>FET has less noise</td></tr>
<tr><td>The frequency changes of BJT will affect its performance</td><td>Its frequency response is high</td></tr>
<tr><td>It depends on the temperature</td><td>Its heat stability is better</td></tr>
<tr><td>It is a low cost</td><td>It is expensive</td></tr>
<tr><td>BJT size is higher as compared with FET</td><td>FET size is low</td></tr>
<tr><td>BJT gain is more</td><td>FET gain is less</td></tr>
<tr><td>Its output impedance is high due to high gain
Difficult to fabricate in ICs</td><td>Its output impedance is low due to low gain
Easy to fabricate in ICs</td></tr>
</table>
</td>
<td>1*7</td>
<td>7</td>
<td>7</td>
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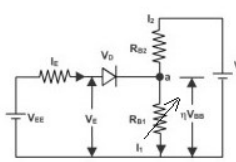
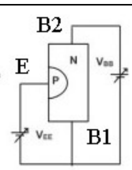
<tr>
<td>X</td>
<td>




Emitter region is heavily doped and close to B2. Inter base Resistance Rbb =Rb2+Rb1, Rb2 resistance from B2 to PN junction, Rb1 junction to B1. When biased as shown, EB1 region forward biased, which injects carriers in lower region reducing Rb1, under forward bias. Voltage across Rb1 reverse biasing PN junction. Until Vee <Vd+Vrb1, junction reverse biased and no emitter current(cut off). When Vee=Vd+Vrb1(peak voltage-Vp), junction forward biased, Ie start flowing, injecting carriers (conductivity modulation), to reduce Rb1. It reduces net reverse bias, increasing Ie even at lower Vee(negative resistance region), until valley voltage, and there after, emitter voltage is almost constant, with increase in Ie(saturation).
</td>
<td>3

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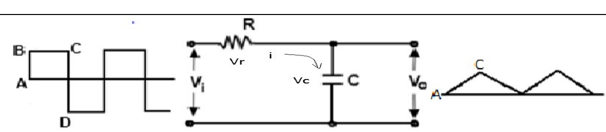
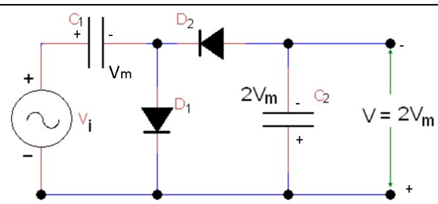
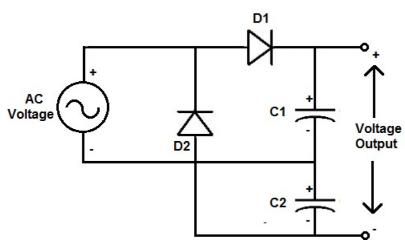
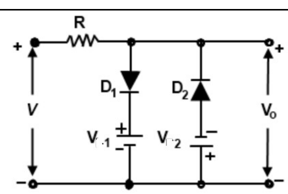
<tr>
<td>XI</td>
<td>


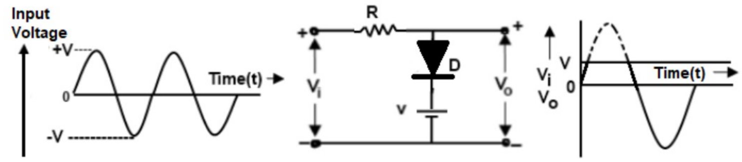
</td>
<td>2</td>
<td>7</td>
<td>7</td>
</tr>
</table>

<table>
<tr><td></td><td>RC integrator circuit. Vi=Vr+Vc. Vo is Vc. i=Vr/r=Vc/Xc.
If Vr>> Vc, Vi=Vr. i=Vi/R.
Vo=Vc; Vc=Q/C; i*t/C; Vi*t/RC. ie output voltage Vo
$= \frac{1}{RC}\int Vi\; dt$; proportional to integral of input
provided R>>Xc, and RC>> time period of input</td><td>5</td><td></td><td></td></tr>
<tr><td>XII</td><td>

Half wave doubler. During +ve half cycle, C1 charges to Vm through D1.during –ve half cycle, C2 charges to 2Vm through D2, C1 and source voltage.


Full wave voltage doubler. +ve half cycle, C1 charges to Vm of i/p through D1. C2 charges to Vm during –ve half cycle. DC o/p voltge across C1 and C2, Vm+Vm= 2Vm.</td><td>1.5
2
1.5
2</td><td>7</td><td>7</td></tr>
<tr><td>XIII</td><td>

Combination diode clipper. During +ve half cycle of i/p, D1 forward biased and D2 reverse biased. D1 not conducting until Vin <V1, and o/p same as i/p. Vin>V1, D1 conducts, o/p =V1. Similar in –ve half cycle.</td><td>4
3</td><td>7</td><td>7</td></tr>
</table>

XIV	 Biased +ve clamper. For Vi<V, diode reverse biased, o/p same as Vi. Vi>v, diode conducts, o/p same as V	4 3	7	7